CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 2: Polynomials

Topic-wise short notes with solved examples

A zero of a polynomial is where its graph meets the x-axis. This chapter links the zeroes of a polynomial to its coefficients.

1 Polynomials and their zeroes

- A **polynomial** in x is an expression such as $p(x) = 3x^2 - 5x + 2$, in which every power of x is a whole number.
- The **degree** is the highest power of x. Degree 1: **linear** ($ax + b$). Degree 2: **quadratic** ($ax^2 + bx + c$). Degree 3: **cubic** ($ax^3 + bx^2 + cx + d$). In each case $a \neq 0$.
- The **value** of $p(x)$ at $x = k$ is $p(k)$, found by putting k in place of x.
- A real number k is a **zero** of $p(x)$ if $p(k) = 0$.
- The zero of the linear polynomial $ax + b$ is $-\frac{b}{a}$.

Example 1.1 Check whether -2 and 3 are zeroes of $p(x) = x^2 - x - 6$.

SOLUTION

$p(-2) = (-2)^2 - (-2) - 6 = 4 + 2 - 6 = 0$

$p(3) = 3^2 - 3 - 6 = 9 - 3 - 6 = 0$

Both -2 and 3 are zeroes of $p(x)$.

2 Zeroes from the graph

- The zeroes of $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.
- **Linear**: the graph is a straight line that cuts the x-axis at exactly one point, so there is exactly one zero.
- **Quadratic**: the graph is a parabola. It opens upwards if $a > 0$ and downwards if $a < 0$.
- A parabola can cut the x-axis at two points (two zeroes), touch it at one point (one zero, counted twice) or not meet it at all (no real zero).
- A polynomial of degree n has **at most** n zeroes. So a quadratic has at most 2 and a cubic has at most 3.

Example 2.1 The graph of $y = p(x)$ cuts the x-axis at three points and does not touch it anywhere else. How many zeroes does $p(x)$ have?

SOLUTION

Each point where the graph meets the x-axis gives one zero.

$p(x)$ has 3 zeroes.

REMEMBER

Count only the points on the x-axis. Points where the graph crosses the y-axis are not zeroes.

3 Zeroes and coefficients of a quadratic polynomial

- If α and β are the zeroes of $ax^2 + bx + c$, then

$$\alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2} \qquad \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

- To find the zeroes, factorise the polynomial by splitting the middle term and put each factor equal to zero.

Example 3.1 Find the zeroes of $x^2 - 2x - 8$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

$x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = (x - 4)(x + 2)$

The zeroes are 4 and -2.

Sum of zeroes $= 4 + (-2) = 2 = -\frac{(-2)}{1} = -\frac{b}{a}$

Product of zeroes $= 4 \times (-2) = -8 = \frac{-8}{1} = \frac{c}{a}$

Zeroes: 4 and -2. The relationship is verified.

Example 3.2 Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship.

SOLUTION

$6x^2 - 7x - 3 = 6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$

The zeroes are $-\frac{1}{3}$ and $\frac{3}{2}$.

Sum $= -\frac{1}{3} + \frac{3}{2} = \frac{7}{6} = -\frac{(-7)}{6}$ and product $= -\frac{1}{3} \times \frac{3}{2} = -\frac{1}{2} = \frac{-3}{6}$

Zeroes: $-\frac{1}{3}$ and $\frac{3}{2}$. The relationship is verified.

Example 3.3 Find the zeroes of $3x^2 - 12$.

SOLUTION

$3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$

Sum $= 2 + (-2) = 0 = -\frac{0}{3}$ and product $= -4 = \frac{-12}{3}$

Zeroes: 2 and -2.

4 Forming a quadratic polynomial

- A quadratic polynomial whose zeroes have sum S and product P is

$$k\,[x^2 - Sx + P], \quad k \neq 0$$

- If S or P is a fraction, choose k to clear the denominators.

Example 4.1 Find a quadratic polynomial whose zeroes have sum -3 and product 2.

SOLUTION

$x^2 - Sx + P = x^2 - (-3)x + 2$

$x^2 + 3x + 2$

Example 4.2 Find a quadratic polynomial whose zeroes have sum $\frac{1}{4}$ and product -1.

SOLUTION

$x^2 - \frac{1}{4}x - 1$. Multiply by 4 to clear the fraction.

$4x^2 - x - 4$

Example 4.3 Find a quadratic polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

SOLUTION

$S = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$

$P = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$

$x^2 - 4x + 1$

5 Expressions in the zeroes

- Never find α and β separately if the question only needs an expression in them. Use $\alpha + \beta$ and $\alpha\beta$.
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$
- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$
- $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Example 5.1 If α and β are the zeroes of $2x^2 - 5x + 3$, find $\alpha^2 + \beta^2$ and $\frac{1}{\alpha} + \frac{1}{\beta}$.

SOLUTION

$\alpha + \beta = \frac{5}{2}$ and $\alpha\beta = \frac{3}{2}$

$\alpha^2 + \beta^2 = \left(\frac{5}{2}\right)^2 - 2 \times \frac{3}{2} = \frac{25}{4} - 3 = \frac{13}{4}$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{5/2}{3/2} = \frac{5}{3}$

$\alpha^2 + \beta^2 = \frac{13}{4}$ and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{5}{3}$

Example 5.2 If one zero of $2x^2 + kx - 15$ is 3, find k and the other zero.

SOLUTION

Since 3 is a zero: $2(3)^2 + 3k - 15 = 0$, so $18 + 3k - 15 = 0$ and $k = -1$.

Product of zeroes $= \frac{c}{a} = -\frac{15}{2}$, so the other zero $= -\frac{15}{2} \div 3 = -\frac{5}{2}$.

$k = -1$ and the other zero is $-\frac{5}{2}$.

6 Zeroes and coefficients of a cubic polynomial

- If α, β and γ are the zeroes of $ax^3 + bx^2 + cx + d$, then

$$\alpha + \beta + \gamma = -\frac{b}{a} \qquad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \qquad \alpha\beta\gamma = -\frac{d}{a}$$

Example 6.1 Verify that 1, 2 and 3 are the zeroes of $x^3 - 6x^2 + 11x - 6$, and check the relationship with the coefficients.

SOLUTION

$p(1) = 1 - 6 + 11 - 6 = 0$, $p(2) = 8 - 24 + 22 - 6 = 0$, $p(3) = 27 - 54 + 33 - 6 = 0$

Sum $= 1 + 2 + 3 = 6 = -\frac{(-6)}{1}$

Sum of products in pairs $= 2 + 6 + 3 = 11 = \frac{11}{1}$

Product $= 6 = -\frac{(-6)}{1}$

All three relations hold.

7 Common mistakes

- Forgetting the minus sign in $\alpha + \beta = -\frac{b}{a}$.
- Reading a, b, c without their signs. In $x^2 - 2x - 8$, $b = -2$ and $c = -8$.
- Writing the polynomial as $x^2 + Sx + P$. The middle term is $-Sx$.
- Saying a quadratic always has two zeroes. It has **at most** two real zeroes.

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